

- 7 Some diseases, and certain cancers, are thought to be caused by a lack of specific neurotransmitters.

This means they can be treated using drugs which increase the concentration of these neurotransmitters.

Some of these drugs are manufactured using genetically modified (GM) organisms.

(a) Which is a description of a GM organism?

(1)

- ☐ **A** an amplified sequence of nucleic acid
- ☐ **B** an organism containing genetic material from another species
- ☐ **C** a virus containing DNA
- ☐ **D** DNA that contains a mutation

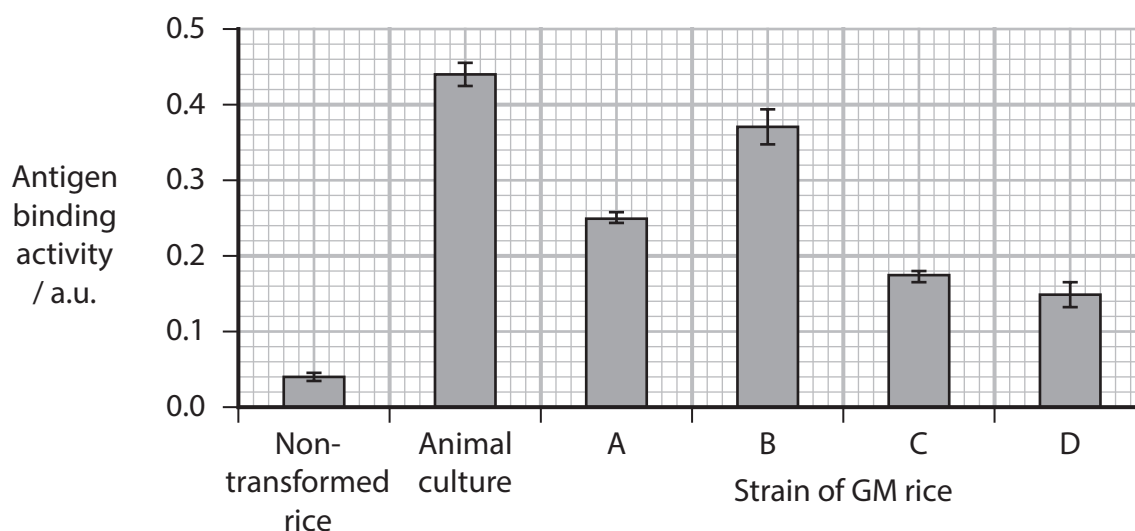
(b) Bevacizumab (BVZ) is an antibody used to treat brain cancer patients.

This antibody is produced using animal cell cultures.

A new method of producing BVZ has been developed using genetically modified (GM) rice.

The graph compares the antigen binding activity of a BVZ antibody produced using:

- a control using non-transformed rice
- animal cell culture
- four strains of GM rice (A, B, C and D).



(i) Describe **three** conclusions that can be made from these results.

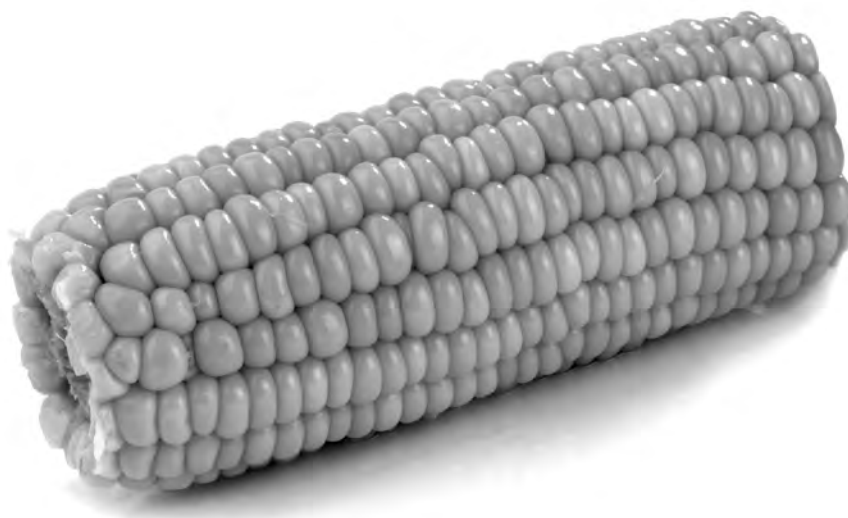
(3)

(ii) Explain how BVZ antibodies can cause the destruction of cancer cells.

(2)

*(c) Maize is a major food crop.

The photograph shows a maize cob with rows of grains.



(Source: © art nick/Shutterstock)

Maize can be infected by a fungus.

This fungus produces a toxin that affects plant growth and is poisonous to animals.

This fungus can be genetically modified (GM).

In an investigation, maize plants were infected with normal fungus or infected with GM fungus.

The table shows the mean and standard deviation for growth measurements of the maize and the concentration of toxin.

Crop infected with	Plant height / cm	Number of grains per cob	Concentration of toxin / a.u.
Normal fungus	285.0 ± 60	113 ± 2	530 ± 100
GM fungus	320.0 ± 33	117 ± 2	270 ± 35

The results are shown in the table.

Use the information given in the tables together with your own knowledge to support your answer.

(6)

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(Total for Question 7 = 12 marks)
